

# INTONATIONAL PHRASING, DISCONTINUITY, AND THE SCOPE OF NEGATION

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*And nothing is, but what is not*

(Macbeth I.iii.142)

*Abstract.* We discuss several cases of English and German negative quantifiers taking extended scope. We argue that these scope extensions are sensitive to linear and prosodic continuity, a fact that we capture in terms of a Condition on Extended Scope Taking (CEST). We provide two formalizations of CEST, one couched in minimalist terms and another within the framework of Combinatory Categorical Grammar (CCG). We compare and contrast the resulting systems and suggest that although the differences are clearly discernible it is too early to judge which of the competitors should be preferred.

## 1. Introduction

Recent influential work in generative syntax, most notably Kayne 1994 and Chomsky 1995, has opened up a way of dispensing with linear notions for “narrowly” syntactic concerns. Linear precedence is accordingly reduced to hierarchical notions via (versions of) Kayne’s Linear Correspondence Axiom (LCA). Combined with the Y-model approach to the linguistic division of labor, linear precedence will play its role on the PF-branch of the grammar exclusively, there being “no clear evidence that order plays a role at LF or in the computation from N to LF” (Chomsky 1995:334).

In this article, however, we would like to discuss some cases of scope taking that *prima facie* contradict this view. We suggest that extending the scope of negative quantifiers in English and German is sensitive to linear and prosodic, that is, PF-properties.<sup>1</sup> This is stated in terms of a Condition on Extended Scope Taking (CEST), which we introduce in section 2.

Section 3 provides two formalizations of CEST. The first is a Y-model preserving minimalist version of Kayne (1998). The crucial ingredient will be a PF-legible feature  $\phi$ , which induces freezing effects in the syntax and continuity effects at PF. This is, of course, reminiscent of the F-feature approach to focus marking as originally introduced by Chomsky (1971). The second one is an extension of Combinatory Categorical Grammar (CCG) (see Steedman 1996, 2000b), which, replacing the Y-model of grammar, directly

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<sup>1</sup> See Bayer 1996 for earlier work in this direction on German, and Vogel and Kenesei 1990 for a closely related study on Hungarian.

encodes conditions on linearity (adjacency, directionality), scope, and intonation in its categories and rules.

We believe that it is too early to decide which of the systems is ultimately more adequate. Instead, we hope that the explicit contrasting of two formalizations within such a narrow domain may help as a future guideline for determining such difficult matters.

## 2. Overt Movement and Extended Scope Taking

The starting point for our discussion is the study of overt versus covert movement by Kayne (1998), where a surface approach to scope taking is advocated. Accordingly,

[i]n a number of cases (involving, e.g., negation, *only*, reverse scope of *some* and *every*, ACD) where covert (LF) phrasal movement has been postulated, it is possible and advantageous to dispense with covert movement (including feature raising) and replace it with a combination of overt movements of phonetically realized phrases. (Kayne 1998:128)

In particular, negative quantifiers, which are our main focus of attention, are taken to be amenable to uniform treatment.

Negative phrases of the (standard) English sort are subject to overt movement to Spec,Neg in all languages. (Kayne 1998:165)

To begin, let us have a closer look at the overt derivation of narrow and wide scope for *not a single linguistics book* in (1).

(1) She has requested that they read not a single linguistics book.

In each case, Neg<sup>0</sup> must be added on top of VP, as illustrated in (2).

- (2) a. [Neg<sup>0</sup> [<sub>VP</sub> read [<sub>DP</sub> not a single linguistics book]]]  
b. [Neg<sup>0</sup> [<sub>VP</sub> requested that they read [<sub>DP</sub> not a single linguistics book]]]

The crucial scoping step consists in overt “Neg-phrase-preposing” into Spec,NegP. This is shown in (3).<sup>2</sup>

<sup>2</sup> It would have to be established whether extraction in (3b) can proceed without intermediate steps. This depends on the theory of locality and (im-)proper movement adopted. However, (i) indicates that more than “just” locality is at stake.

(i) John believes that they read not a single linguistics book.

Thus, although complements of *believe* are transparent for extraction, wide-scope construal of *not a single linguistics book* is unavailable. It is also interesting to note that despite the fact that *believe* (or *glauben* in German) is a typical context allowing for so-called neg-raising, which used to be analyzed as “a lexically governed, cyclic, structure-preserving rule which extracts a negative element from a lower clause...and raises it one clause up...over a predicate marked to allow the rule’s application” (Horn 1989:312) (cf. Fillmore’s [1963] (ii), quoted from Horn [1989:313]), negation does not seem to be able to take (inverse) scope over the universal quantifier in (iii) as would be required for so-called bridge-contour-induced interpretation. According to Büring (1996: 112), “although judgements are hard to make in these cases...this sentence with the accent pattern indicated cannot be interpreted, i.e. is understood to be weird, a slip of the tongue or the like.”

- (3) a. [<sub>NEGP</sub> [<sub>DP</sub> not a single linguistics book]<sub>i</sub> [<sub>Neg<sup>0</sup></sub> [<sub>VP</sub> read *t<sub>i</sub>*]]]  
 b. [<sub>NEGP</sub> [<sub>DP</sub> not a single linguistics book]<sub>i</sub> [<sub>Neg<sup>0</sup></sub> [<sub>VP</sub> requested that they read *t<sub>i</sub>*]]]

On the assumption that, in consonance with the LCA (Kayne 1994), specifiers invariably precede their sister constituents, an additional step is required for restoring surface word order, at least in VO languages like English. The landing site for this is provided by WP, a phrase whose head *W<sup>0</sup>* attracts *Neg<sup>0</sup>* and checks [+w] features (mnemonic for “word order”) against VP in Spec,WP. The result of “*Neg<sup>0</sup>*-raising” and “*VP*-preposing” is given in (4).

- (4) a. [<sub>WP</sub> [<sub>VP</sub> read *t<sub>i</sub>*]<sub>j</sub> [<sub>Neg<sup>0</sup></sub><sub>k</sub>-*W<sup>0</sup>* [<sub>NEGP</sub> [<sub>DP</sub> not a single linguistics book]<sub>i</sub> [*t<sub>k</sub>* *t<sub>j</sub>*]]]]]  
 b. [<sub>WP</sub> [<sub>VP</sub> requested that they read *t<sub>i</sub>*]<sub>j</sub> [<sub>Neg<sup>0</sup></sub><sub>k</sub>-*W<sup>0</sup>* [<sub>NEGP</sub> [<sub>DP</sub> not a single linguistics book]<sub>i</sub> [*t<sub>k</sub>* *t<sub>j</sub>*]]]]]

Given the different c-command domains of the negative quantifiers, it can be postulated that (3a) and (3b) are crucial steps in deriving narrow- and wide-scope readings, such as sketched in (5a) and (5b), respectively.<sup>3</sup>

- (5) a. REQUESTED(she,  $\neg\exists x[\text{LB}(x) \wedge \text{READ}(\text{they}, x)]$ )  
 b.  $\neg\exists x[\text{LB}(x) \wedge \text{REQUESTED}(\text{she}, \text{READ}(\text{they}, x))]$

The same mechanism applies *mutatis mutandis* to (6), again yielding either narrow or wide scope for the negative quantifier *no one*.

- (6) They have requested that we turn down no one.

One of the interesting challenges facing this approach is constituted by examples like the ones in (7), where wide-scope construal of the negative quantifier is systematically degraded.

- (7) a. She has requested that not a single student read our book.  
 b. They have requested that we turn no one down.

- (ii) a. I don't believe that he wants me to think that he did it.  
 b. [\_\_\_\_ I believe [\_\_\_\_ he wants [\_\_\_\_ I think [NOT he did it]]]]  
 (iii) ??/ALLE Politiker glauben, daß du es NICHT\ schaffst.  
 all politicians believe that you it not make  
 'All politicians believe that you will not succeed.'

<sup>3</sup> We sidestep the issue of intensionality and its effect on potential additional readings. See de Swart 2000 for a recent semantic analysis of negative quantifiers.

The analysis of wide-scope negation in (7) would, according to Kayne (1998), require an initial insertion of a predication head  $\text{Pred}^0$  on top of VP. (8) illustrates this.

- (8) a.  $[\text{Pred}^0 [\text{VP requested that } [\text{DP not a single student}] [\text{VP read our book}]]]$   
 b.  $[\text{Pred}^0 [\text{VP requested that we turn } [\text{DP no one}] \text{ down}]]]$

As shown in (9), this would be followed by “VP-preposing,” (9a), or “long-particle-preposing,” (9b), into Spec,PredP.

- (9) a.  $[\text{PredP } [\text{VP read our book}]_i [\text{Pred}^0 [\text{VP requested that } [\text{DP not a single student}] t_i]]]$   
 b.  $[\text{PredP } [\text{down}]_i [\text{Pred}^0 [\text{VP requested that we turn } [\text{DP no one}] t_i]]]$

From here on, everything would proceed as before—that is, via a sequence of “Neg<sup>0</sup>-insertion,” “Neg-phrase-preposing,” “W<sup>0</sup>-insertion,” “Neg<sup>0</sup>-raising,” and “VP-preposing.” (10) and (11) illustrate two essential stages of this derivation.

- (10) a.  $[\text{NegP } [\text{DP not a single student}]_j [\text{Neg}^0 [\text{PredP } [\text{VP read our book}]_i [\text{Pred}^0 [\text{VP requested that } t_j t_i]]]]]$   
 b.  $[\text{NegP } [\text{DP no one}]_j [\text{Neg}^0 [\text{PredP } [\text{VP down}]_i [\text{Pred}^0 [\text{VP requested that we turn } t_j t_i]]]]]$
- (11) a.  $[\text{WP } [\text{VP requested that } t_j t_i]_i [\text{Neg}^0_{\text{k-W}^0} [\text{NegP } [\text{DP not a single student}]_j [t_k [\text{PredP } [\text{VP read our book}]_i [\text{Pred}^0 t_i]]]]]]]$   
 b.  $[\text{WP } [\text{VP requested that we turn } t_j t_i]_i [\text{Neg}^0_{\text{k-W}^0} [\text{NegP } [\text{DP no one}]_j [t_k [\text{PredP } [\text{VP down}]_i [\text{Pred}^0 t_i]]]]]]]$

To the extent that the facts in (7)–(11) are as parallel as they appear from our presentation, searching for a uniform way of ruling out these derivations may seem attractive. We will ourselves conduct such a search below. Let us note first, however, that Kayne (1998) envisages distinct approaches to (7a) and (7b). As for the latter:

We conclude, therefore, that the deviance of wide scope in [(7b)] must be due to the long-distance particle preposing seen in the first step of [(9b)]. (Kayne 1998:143)

Unfortunately, no detailed further account is given. In fact, the two ways of ruling out “long-distance particle preposing” (LDPP) that come to mind most readily are not straightforward. Thus, assume that  $\text{Pred}^0$ , governing the VP headed by *requested*, does not provide a landing site for *down*, due to lack of lexical licensing.<sup>4</sup> This would distinguish (9b) from (12), the latter taken by

<sup>4</sup> See Hinterhölzl 1999 for such an assumption with respect to Dutch and German counterparts of (7b).

Kayne to be a necessary step in deriving the well-formed narrow-scope construal of (7b).

- (12) [<sub>PredP</sub> [down]<sub>i</sub> [<sub>Pred</sub><sup>0</sup> [<sub>VP</sub> turn [<sub>DP</sub> no one] *t<sub>i</sub>*]]]

It would, however, have to be additionally shown that no WP could do the job of PredP instead—that is, a theory of “word order phrases” would have to be developed. Doing this in a nonstipulative way may be nontrivial.<sup>5</sup>

A second approach might appeal to locality as the crucial factor. Thus, finite (subjunctive) CP might be taken to be a barrier for LDPP.<sup>6</sup> However, this approach, while feasible in the case of (7b), may not be sufficient, given that the contrast between (6) and (7b) can be replicated in terms of control infinitives. This is shown in (13).

- (13) a. They have forced us to turn down no one.  
b. They have forced us to turn no one down.

Again, only (13a) possesses an additional wide-scope reading of the negative quantifier *no one*. The problem posed by (13) is that declaring control infinitives opaque for LDPP would have to be a stipulated special property of English, given the well-formedness of Dutch (14).

- (14) omdat hij [mij]<sub>i</sub> [op]<sub>j</sub> probeert [te bellen *t<sub>i</sub>* *t<sub>j</sub>*]  
because he me up tries to call  
'because he tries to call me up'

<sup>5</sup> The same reasoning applies *mutatis mutandis* to (9a). A principled approach to the distribution of WP could adopt Müller's (2002) concept of “repair-driven movement.” This global last resort mechanism licenses non-feature-driven movement in order to restore basic word-order relations. Thus, English VO order, broken up by “Neg-phrase preposing,” would be restored by VP-preposing into Spec,WP. On a cyclic, bottom-up version of such a theory, non-feature-driven movement of VP and particle *down* in (i) can be prevented, because word order is still canonical at that stage.

- (i) a. \*<sub>[WP</sub> [<sub>VP</sub> read our book]<sub>i</sub> [<sub>W</sub><sup>0</sup> [<sub>VP</sub> requested that [not a single student] *t<sub>i</sub>*]]]  
b. \*<sub>[WP</sub> [down]<sub>i</sub> [<sub>W</sub><sup>0</sup> [<sub>VP</sub> requested that we turn [no one] *t<sub>i</sub>*]]]

What makes this approach less attractive in the current context is the fact that feature-driven movement types have to be put into two distinct classes. The class containing “Neg-phrase preposing” would invite repair-driven movement, whereas topicalization, for example, would disallow such a process. Thus, repair-driven “IP-fronting,” as illustrated in (iib), has to be prevented from obligatorily masking structures like (iia).

- (ii) a. [[This point]<sub>i</sub> [<sub>IP</sub> we have to concede *t<sub>i</sub>*]]  
b. [<sub>WP</sub> [<sub>IP</sub> We have to concede *t<sub>i</sub>*]<sub>j</sub> [<sub>W</sub><sup>0</sup> [[this point]<sub>i</sub> *t<sub>j</sub>*]]]

Clearly, such a systematic distinction between movement types comes close to restoring the distinction between overt versus covert movement, which Kayne (1998) set out to dismantle.

<sup>6</sup> The notions of barrier and movement type would have to be fine-grained enough, of course, for allowing “Neg-phrase-preposing” out of such complements. Kayne (1998:137) mentions the possibility of taking “particle preposing” to be X<sup>0</sup>-movement. This would seem to allow the HMC to rule out (9b). However, without extra assumptions even narrow-scope negation in (7b) would be ruled out by the HMC, given that *down* has to be extracted from VP in this case as well.

In the context of Kayne (1998) this issue is further complicated by the fact that even “short particle preposing” has to be somehow constrained, given the contrast in (15).

- (15) a. John pointed out Bill, too.  
b. ??John pointed Bill, too, out.

Dealing with the relation between *too* and its semantic associate in terms of overt “attraction to *too*” is an additional objective of Kayne (1998). (16) and (17) illustrate the crucial derivational steps for (15a) and (15b), respectively.

- (16) a. [too [<sub>VP</sub> pointed Bill out]]  
b. [[<sub>DP</sub> Bill]<sub>i</sub> [too [<sub>VP</sub> pointed *t<sub>i</sub>* out]]]  
c. [<sub>WP</sub> [<sub>VP</sub> pointed *t<sub>i</sub>* out]<sub>j</sub> [<sup>W<sup>0</sup></sup> [[<sub>DP</sub> Bill]<sub>i</sub> [too *t<sub>j</sub>*]]]]]
- (17) a. [<sup>Pred<sup>0</sup></sup> [<sub>VP</sub> pointed Bill out]]  
b. [<sub>PredP</sub> [out]<sub>i</sub> [<sup>Pred<sup>0</sup></sup> [<sub>VP</sub> pointed Bill *t<sub>i</sub>*]]]  
c. [too [<sub>PredP</sub> [out]<sub>i</sub> [<sup>Pred<sup>0</sup></sup> [<sub>VP</sub> pointed Bill *t<sub>i</sub>*]]]]]  
d. [[<sub>DP</sub> Bill]<sub>j</sub> [too [<sub>PredP</sub> [out]<sub>i</sub> [<sup>Pred<sup>0</sup></sup> [<sub>VP</sub> pointed *t<sub>j</sub>* *t<sub>i</sub>*]]]]]  
e. [<sub>WP</sub> [<sub>VP</sub> pointed *t<sub>j</sub>* *t<sub>i</sub>*]<sub>k</sub> [<sup>W<sup>0</sup></sup> [[<sub>DP</sub> Bill]<sub>j</sub> [too [<sub>PredP</sub> [out]<sub>i</sub> [<sup>Pred<sup>0</sup></sup> *t<sub>k</sub>*]]]]]]]

Clearly, neither landing site nor locality of “short particle preposing” seem to be responsible for the degraded status of (15b).<sup>7</sup>

Let us next turn to the account of (7a) suggested in Kayne 1998.

We can maintain the idea that the deviance of the wide-scope reading in [(7a)] is closely related to the deviance found in many well-known cases of overt extraction from preverbal subject position (e.g., in *wh*-constructions of various types). (Kayne 1998:144)

Again, this idea is not elaborated on more fully.

I will set aside the (important) question of whether it is the ECP that is at issue, or some other (kind of) constraint. (Kayne 1998:145, fn. 38)

In fact, it is not immediately obvious how to formulate the required extraction related constraint. Thus, consider (18).

- (18) a. Not a single student did she request that—given lack of time—  
read our book in its entirety.  
b. Not a single linguistics book did she request that—given lack of  
time—we read in its entirety.

As pointed out by Culicover (1993), insertion of an adjunct immediately to the right of complementizer *that* suspends a *that*-trace effect. Consequently, (18a) and (18b) are equally acceptable. However, enhanced extractability does not

<sup>7</sup> Although we are not going to treat “association with *too*,” the account of negative quantifier scope below is relevant for the facts in (15) as well.

result in a wide-scope option for the negative quantifier in subject position. Thus, (19) does not differ from (7a) in that respect.

- (19) She requested that—given lack of time—not a single student read our book in its entirety.

We take these apparent difficulties in accounting for the contrasts in (1), (6), and (13a) versus (7a), (7b), and (13b) to justify an attempt at developing an alternative. A look at surface word order reveals that the negative quantifiers able to take wide scope are all located at the right edge in string-final position. In other words, the region these quantifiers extend their scope over is linearly continuous. This is schematically shown in (20).<sup>8</sup>

- (20) ( $\sigma$  .....):(Q<sub>-</sub>)

Examples (1), (6), and (13a), cast in this format, are given in (21).<sup>9</sup>

- (21) a. She has ( $\sigma$  requested that they read):(Q<sub>-</sub> not a single linguistics book)  
 b. They have ( $\sigma$  requested that we turn down):(Q<sub>-</sub> no one)  
 c. They have ( $\sigma$  forced us to turn down):(Q<sub>-</sub> no one)

By contrast, the structures that disallow wide-scope negation show discontinuous “ $\sigma$ -regions.” This is schematically illustrated in (22).

- (22) ( $\sigma$  ...):(Q<sub>-</sub>):( $\sigma$  ...)

Examples (7a), (7b), and (13b) are mapped onto this schema in (23).

- (23) a. She has ( $\sigma$  requested that):(Q<sub>-</sub> not a single student):( $\sigma$  read our book)  
 b. They have ( $\sigma$  requested that we turn):(Q<sub>-</sub> no one):( $\sigma$  down)  
 c. They have ( $\sigma$  forced us to turn):(Q<sub>-</sub> no one):( $\sigma$  down)

Interestingly, linear continuity of the  $\sigma$ -region is not a sufficient condition for negative quantifier scope to be extendable. Thus, consider the effect of inserting an intonational phrase (IP<sup>0</sup>)-boundary (||) in example (1), as shown in (24).<sup>10</sup>

<sup>8</sup> A closely related linear effect on quantifier scope has been observed by Williams (1986:272), as the following quote shows. “It may be that wide-scope readings are best if the quantifier is in final position, which is the natural ‘focus’ position.” However, we doubt that (narrow) focus is directly involved here. Thus, the negative quantifier of (7b) does not seem to acquire a wide scope reading if focused. The relevant example is given in (i).

(i) They have requested that we turn NO ONE down.

<sup>9</sup> Our decision on the left edge of  $\sigma$  is modeled after the exposition in Kayne 1998 and would have to be subjected to further analysis in terms of the principles introduced below.

<sup>10</sup> We somewhat impressionistically distinguish intonational phrase boundaries (||) from intermediate phrase boundaries (|) (cf. Ladd 1996).

(24) She has requested || that they read not a single linguistics book.

The wide-scope option is lost here too, which may be surprising from a purely syntactic point of view. However, the IP<sup>Q</sup>-boundary interferes with the continuity of the  $\sigma$ -region as well. Let us therefore make the following descriptive generalization.<sup>11,12</sup>

<sup>11</sup> Similar facts can be observed in Italian, although the situation is slightly more complicated there, due to a different status of negative phrases in that language. In Italian, as in many other Romance languages, negative phrases—depending on their position in the clause—show behavior of negative quantifiers (in preverbal, or clause-initial, position) or they behave like NPIs (in postverbal, or non-clause-initial, position), as documented by, among others, Rizzi (1982), Laka (1990), Zanuttini (1994), Haegeman (1995), Tovena (1996), Herburger (1998), and Giannakidou (1998). Nevertheless, in Italian, just as in English, the extended scope taking of negative phrases seems to be subject to the condition postulated in (25) (see also Steedman 1996:57). In (ia), the negative phrase in postverbal position must be constructed with the matrix negation (“take wide scope”) to be grammatical. In contrast, the negative phrase in preverbal position in (ib) cannot do so, thus giving rise to a double-negation reading, as indicated in the translation. Note that only in (ia), but not (ib), the requirements of the CEST in (25) are met: the region  $\sigma$  over which the negative phrase takes scope is linearly (and prosodically, see below) continuous; cf. (ia) versus (ib), respectively.

- (i) a. Non voglio che venga nessuno.  
not want-1SG that come no one  
‘I don’t want anyone to come.’  
b. Non voglio che nessuno venga.  
not want-1SG that no one come  
‘I don’t want that no one come(s).’  
(ii) a. ( $\sigma$  Non voglio che venga):( $Q_{-}$ , nessuno)  
b. ( $\sigma$  Non voglio che):( $Q_{-}$ , nessuno):( $\sigma$  venga)

That indeed prosodic continuity or the lack thereof is at issue in examples (ia) and (ib) is demonstrated in (iii). (iiia) shows that inserting an IP-boundary into (ia) leads to ungrammaticality because the negative phrase cannot be constructed with the matrix negation any longer. Negation, however, is needed for the licensing of a postverbal negative phrase, due to its NPI status. In contrast, an IP-boundary is in fact required for (ib) to be grammatical. According to Rizzi (1982:174, fn. 12), “such sentences as [(ib)] are only acceptable with a peculiar intonational pattern, i.e., with main stress on the main verb, and an intonational break between the verb and the sentential complement.” Compare (iiib).

- (iii) a. ??Non voglio || che venga nessuno.  
b. Non VOGLIO || che nessuno venga.

<sup>12</sup> Given CEST, examples like (i), from Longobardi (1991:171), seem to pose a problem for our analysis. In (i), the negative quantifier is embedded into a larger NP. Nevertheless, it can take wide scope out of this larger NP. Note that the region  $\sigma$  over which the negative quantifier takes scope in (i) is discontinuous as shown in (ii).

- (i) (?)La presenza di nessuno lo spaventerebbe.  
‘The presence of no one would frighten him.’  
(ii) ( $\sigma$ La presenza di):( $Q_{-}$ , nessuno):( $\sigma$ lo spaventerebbe)

To account for the wide scope of negative phrases in cases like (i), Longobardi proposes a percolation mechanism: the whole (large) NP counts as quantified due to an upward percolation of a [+Q] feature from the negative phrase. It seems to us that examples like (i) could in fact be made compatible with our approach by assuming that it is the whole (“large”) NP that counts for establishing the relevant intonational domain (Daniel Büring, p.c.).



(25) *Condition on Extended Scope Taking (CEST)*

Extending the scope of a negative quantifier  $Q\neg$  over a region  $\sigma$  requires  $\sigma$  to be linearly and prosodically continuous.

Translating this back into the perspective of Kayne (1998), we suggest that instead of seeking a ban on “VP-preposing” or LDPP in cases like (7) one could view CEST, or whatever eventually underlies it, as a filter on the output of syntax. Thus, to give just one example, reconsider the structures in (4b) and (11a), repeated as (26a) and (26b) respectively.

- (26) a.  $[_{WP} [_{VP} \text{requested that they read } t_i]_j [_{Neg^0_k-W^0} [_{NegP} [_{DP} \text{not a single linguistics book}]_i [t_k \ t_j]]]]]$   
 b.  $[_{WP} [_{VP} \text{requested that } t_j \ t_i]_l [_{Neg^0_k-W^0} [_{NegP} [_{DP} \text{not a single student}]_j [t_k \ [_{PredP} [_{VP} \text{read our book}]_i [Pred^0 \ t_l]]]]]]]]]$

(26a) would allow a one-to-one mapping between Spec,WP and the  $\sigma$ -region of (25), yielding the continuous pattern in (20). (26b), on the other hand, requires both Spec,WP and Spec,PredP to be mapped into  $\sigma$ , yielding the discontinuous pattern (22).

Let us next inquire further into the factors underlying CEST. The following example confirms the importance of prosodic continuity for  $\sigma$ .

- (27) She requested that the students who finish first read not a single linguistics book.

Replacing the phonologically “weak” personal pronoun *they* by a “heavy” subject interferes with wide-scope construal of the negative quantifier. The crucial effect on intonational phrasing here is that at least an intermediate phrase (ip<sup>0</sup>)-boundary (|) has to intervene between *first* and *read*. This is shown in (28).

- (28) She requested that the students who finish first | read not a single linguistics book.

(28), of course, displays a prosodically discontinuous  $\sigma$ -region in violation of CEST. It is not immediately obvious how the contrast between (1) and (27) could be insightfully captured in purely syntactic terms.

Note also that CEST must be a condition on *extending* scope, not “standard” scope. Thus, of course, narrow scope of the negative quantifier in (7b) is compatible with the discontinuous phrasal verb. This interpretation of CEST yields an interesting prediction for ECM constructions.<sup>13</sup> Consider (29).

- (29) We expect not a single student to have read this book.

<sup>13</sup> Thanks to Chris Wilder (p.c.) for raising this issue.

Clearly, the negative quantifier can take scope over *expect* in (29). At the same time, (29) would be an instantiation of the discontinuous pattern in (22), as (30) shows.

(30) We ( $\sigma$  expect):( $Q_{-}$  not a single student):( $\sigma$  to have read this book)

We therefore suggest that (29) is not an instance of extended scope taking. Instead it would be evidence for some version of “raising to object,” which allows the negative quantifier to be a constituent of the matrix clause, the latter being the domain of its “standard” scope.<sup>14</sup>

Kayne (1998) also includes some discussion of facts from German, so it is interesting to see how CEST fares in this domain. Consider first example (31), which lacks a wide-scope option for the negative quantifier *niemanden* ‘nobody’.

(31) dass sie versprach niemanden zu grüßen  
that she promised nobody to greet

This has been explicitly noted by Kayne (1998:175):

Thus [(31)] differs from the similar English examples, for which the wide scope reading is available.

Given that *niemanden* is a direct object, prospects for a locality or ECP-based account are even bleaker for (31) than they were for the contrast between (1) and (7a) discussed previously.<sup>15</sup> The acceptability of (32) would seem to provide evidence against such an account.

(32) dass sie niemanden<sub>i</sub> versprach [<sub>i</sub> zu grüßen]

These German facts may appear surprising under the premises of Kayne 1994, according to which all languages underlyingly show head < complement orders. However, it is helpful to take into account the OV property distinguishing German from English VO at least at the surface. Thus, the infinitive in (31) is traditionally taken to be “extraposed.” This correlates with the fact that an IP<sup>0</sup>-boundary has to precede it, as shown in (33).<sup>16</sup>

(33) dass sie versprach || niemanden zu grüßen

<sup>14</sup> For evidence from intonational phrasing of ECM structures that seems to count in favor of our hypothesis, see Taglicht 1998. See Lasnik 1995, 1999 for a more recent defense of a syntactic analysis compatible with our claim. To account for the narrow-scope reading of the negative quantifier in (30), one could assume that in this case it is the lower copy of *not a single student* that is chosen for interpretation at LF. See Fox 1999 for pertinent discussion.

<sup>15</sup> Kayne (1998:176, fn. 109) makes some remarks in the same direction.

<sup>16</sup> See Truckenbrodt 1995 for prosodic analysis of extraposition in German. See also Bech 1955 for remarks on the relevance of intonational boundaries in German infinitival constructions. See Haegeman and van Riemsdijk 1986, Koster 1987, and Bayer 1996 for earlier treatments of the interaction between directionality and scope.

Thus, from the perspective of CEST, the missing wide-scope option of (31)/(33) is in fact expected, given that the  $\sigma$ -region *niemanden* is supposed to scope over is neither linearly nor prosodically continuous. Conversely, a wide-scope option arises as soon as the infinitive is “intraposed,” as shown in (34).

(34) dass sie niemanden zu grüßen versprach

In keeping with CEST, scope of *niemanden* correlates with intonational phrasing. Thus, wide scope is possible if an  $\text{ip}^\circ$ -boundary immediately precedes *zu grüßen*, (35a), whereas *niemanden* is confined to narrow scope if such a boundary separates *zu grüßen* and *versprach*, (35b).

- (35) a. dass sie niemanden | zu grüßen versprach  
b. dass sie niemanden zu grüßen | versprach

(36a–c) provide the CEST-relevant patterns for (31), (35a), and (35b), respectively.

- (36) a.  $(\sigma \text{ versprach}) : (\neg, \text{niemanden}) : (\sigma \text{ zu grüßen})$   
b.  $(\neg, \text{niemanden}) : (\sigma \text{ zu grüßen versprach})$   
c.  $(\neg, \text{niemanden}) : (\sigma \text{ zu grüßen}) (\sigma \text{ versprach})$

Again CEST makes the right predictions, whereas attempting to derive these patterns in terms of extractability is not straightforward. We have already seen in (32) that independently extraction of *niemanden* from the extraposed infinitival structure is possible. Thus, the remaining option for ruling out the wide-scope construal of (31) in terms of movement theory lies in restricting the extraction of *zu grüßen*. However, (37) shows that this constituent can be moved independently too.

(37) [Zu grüßen] hat sie niemanden versprochen.

What’s more, such an independent movement step seems to be required in order to derive the wide-scope reading of *niemanden* in (34). This derivation can be modeled on the derivation of (7a) via (8a), (9a), and (10a).<sup>17</sup>

- (38) a.  $[\text{Pred}^0 [\text{VP versprach} [[\text{DP niemanden}]_i [\text{VP zu grüßen } t_i]]]$   
b.  $[\text{PredP} [\text{VP zu grüßen } t_i]_j [\text{Pred}^0 [\text{VP versprach} [[\text{DP niemanden}]_i t_j]]]$   
c.  $[\text{NegP} [\text{DP niemanden}]_i [\text{Neg}^0 [\text{PredP} [\text{VP zu grüßen } t_i]_j [\text{Pred}^0 [\text{VP versprach } [t'_i t_j]]]]]]]$

Due to surface OV order in German, no further VP-to-Spec,WP step is required here, in contrast to what is happening in (11a). However, there is no

<sup>17</sup> Assume that *niemanden* has to undergo a short licensing step inside of the infinitive, for example, into Spec,Agr<sub>OP</sub>. This would be sufficient for inducing OV order in (31).

reason why such a step should be ruled out under the perspective of Kayne (1998). Unfortunately, such a step, as depicted in (39), would yield the basis for deriving the unwelcome wide-scope reading for (31).

- (39)  $[_{WP} [_{VP} \text{versprach } [t'_i t_j]]]_i [Neg^0_k -W^0 [_{NegP} [_{DP} \text{niemanden}]_i [t_k [_{PredP} [_{VP} \text{zu grüßen } t_i]_j [Pred^0 t_i]]]]]]]$

Further evidence that a CEST-based approach may be appropriate also for German is provided by (40).

- (40) weil sie [niemanden zu grüßen] | regelmäßig versprach  
 because she no one to greet regularly promised  
 'because she promised regularly not to greet anybody'

The adverbial *regelmäßig*, modifying the matrix verb, induces an  $ip^\phi$ -boundary on its left. Once more the wide scope option for the negative quantifier is lost. The CEST-relevant structuring, displaying a prosodically discontinuous  $\sigma$ -region, is given in (41).

- (41)  $(Q_{-} \text{niemanden}) : (\sigma \text{ zu grüßen}) (\sigma \text{ regelmäßig versprach})$

In sum, we have diagnosed a systematic gap in the surface approach to scoping of Kayne (1998). We believe that CEST, as formulated in (25) above, constrains the way negative quantifiers scope out of infinitival and subjunctive complements in English and German.<sup>18</sup> We therefore suggest that the capability of implementing CEST should be used as an adequacy criterion for accounts of the above scoping facts.

In the following, two such implementations are going to be provided and discussed. The first one is a Y-model preserving minimalist analysis that introduces CEST quite directly into the system of Kayne (1998). Subsequently, we look at a system that incorporates notions of adjacency and linear

<sup>18</sup> We have to stress that our coverage of German is quite limited. An anonymous reviewer rightly points out that, as (i) indicates, CEST alone may not be a sufficient condition for German.

(i) Sie meinte er liebe niemanden.  
 she thought he like.SUBJ no one  
 'She thought he loved no one.'

(ii) gives a CEST-compatible phrasing for (i).

(ii)  $(\sigma \text{ Sie meinte er liebe}) : (Q_{-} \text{niemanden})$

Nevertheless, the wide-scope option for *niemanden* is missing. The key to an analysis of (i) appears to lie in taking seriously the canonical selectional directionality of German verbs (right to left) and the fact that V2-complements have to be extraposed obligatorily; that is, they show up on the "wrong" side (cf. Stechow & Sternefeld 1988). See Bayer 1996 for a wealth of scoping facts confirming these factors. See section 3.2 for a more thorough consideration of directionality.

order more directly—namely, the categorial approach by Steedman (1996, 2000b).<sup>19</sup>

### 3. Two Implementations of CEST

#### 3.1 Overt Movement

Assume that we keep the system of Kayne (1998) as introduced previously. For the implementation of CEST we further assume that each clause, finite or infinitival, contains a projection of  $\text{Pol}^0$ —that is, a functional head determining the polarity of the clause.<sup>20</sup>  $\text{Pol}^0$  comes in three varieties—namely,  $-\text{Pol}^0$ ,  $+\text{Pol}^0$ , and  $\%\text{Pol}^0$ —where the first replaces  $\text{Neg}^0$ , the second is used in positive emphatic (or VERUM-focused) structures, and the underspecified  $\%\text{Pol}^0$  is used elsewhere.  $\%\text{Pol}^0$  introduces a weak underspecified feature  $[\%neg]$ . This feature does not attract anything by itself. However, it serves as an intervener causing minimality violations—in other words, no polarity dependency can be established across  $\%\text{Pol}^0$  as long as  $[\%neg]$  is unchecked. At the same time, unchecked  $[\%neg]$ -features do not lead to any interface violations. They can be erased by the Spell-Out mechanism.

Now, crucially there have to be two variants of  $-\text{Pol}^0$ , the standard one,  $-\text{Pol1}^0$ , carries a strong XP-feature  $[+neg]$  which can be checked by moving a negative quantifier into  $\text{Spec,PolP}$ . We find  $-\text{Pol1}^0$  in local contexts like (42).

- (42) a. We turned no one down.  
 b.  $[_{\text{WP}} [_{\text{VP}} \text{turned } t_i] [_{\text{W}^0} [_{\text{PolP}} [_{\text{DP}} \text{no one}]_i [-\text{Pol1}^0 [_{\text{PredP}} [\text{down}]]]]]]]$

In nonlocal contexts such as (43), however,  $[\%neg]$  on  $\%\text{Pol}^0$  will block attraction of the quantifier by  $-\text{Pol1}^0$ .<sup>21</sup>

- (43)  $[_{\text{PolP}} -\text{Pol1}^0_{[+neg]} [_{\text{VP}} \text{requested that we } [_{\text{PolP}} \%\text{Pol}^0_{[\%neg]} [_{\text{VP}} \text{turn down no one}]]]]]$

$[+neg]$  on  $-\text{Pol1}^0$ , as well as on *no one*, remains unchecked. Consequently, (43) does not possess any well-formed continuation.

The second variant of  $-\text{Pol}^0$  (i.e.,  $-\text{Pol2}^0$ ) is responsible for CEST-compatible scope extensions. This functional head must possess the following properties.

<sup>19</sup> It would have been interesting to equally reconsider the reanalysis approach by Haegeman and van Riemsdijk (1986). We have to leave this for further research.

<sup>20</sup> Variants of Polarity Phrase (PolP) have been assumed by, among others, Pollock (1989:421, fn. 51) (“AssertionP”) and Laka (1990) (“SigmaP”).

<sup>21</sup> One could think of  $\%\text{Pol}^0$  as a “defective probe” in the sense of Chomsky 2000.

(44) *Properties of  $-\text{Pol}2^0$* 

- a.  $-\text{Pol}2^0$  checks  $[\%neg]$  on  $\%Pol^0$ .
- b.  $-\text{Pol}2^0$  checks  $[+neg]$  on a negative quantifier  $Q \neg$  attracting  $Q \neg$  into Spec,PolP.
- c.  $-\text{Pol}2^0$  assigns a feature  $\varphi$  to its sister constituent XP, such that
  - (i) XP is turned into an island for extraction, and
  - (ii) insertion of a prosodic boundary into XP is prevented.

(45) illustrates the functioning of  $-\text{Pol}2^0$ .

- (45) a.  $[\text{PolP } -\text{Pol}2^0_{[+neg]} [\text{VP requested that we } [\text{PolP } \%Pol^0_{[\%neg]} \text{ turn down } [\text{DP no one}]_{[+neg]}]]]$   
*Check  $[\%neg]$*
- b.  $[\text{PolP } -\text{Pol}2^0_{[+neg]} [\text{VP requested that we } [\text{PolP } \%Pol^0_{[\%neg]} \text{ turn down } [\text{DP no one}]_{[+neg]}]]]$   
*Check  $[+neg]$ , attract DP*
- c.  $[\text{PolP } [\text{DP no one}]_{[+neg]} [-\text{Pol}2^0_{[+neg]} [\text{VP requested that we } [\text{PolP } \%Pol^0_{[\%neg]} \text{ turn down } t_i]]]]]$   
*Assign  $\varphi$*
- d.  $[\text{PolP } [\text{DP no one}]_{[+neg]} [-\text{Pol}2^0_{[+neg]} [\text{VP}[\varphi] \text{ requested that we } [\text{PolP } \%Pol^0_{[\%neg]} \text{ turn down } t_i]]]]]$

Given our stipulation about  $\varphi$  in (44c), we can now derive the crucial contrasts from section 1. Thus, (45) can properly be continued by raising  $\text{VP}[\varphi]$  into Spec,WP. Ultimately, this structure converges as (6) with a wide-scope interpretation for *no one*. Likewise, the interface condition (44cii) prevents the insertion of prosodic boundaries into  $\text{VP}[\varphi]$ , thereby ruling out (24) and (27)/(28) at PF. Finally, the freezing condition (44ci) is responsible for the ill-formedness of a wide-scope reading of (7a,b). Consider (7b) at the stage where  $\varphi$  has been assigned. This is shown in (46).

- (46)  $[\text{PolP } [\text{DP no one}]_{[+neg]} [-\text{Pol}2^0_{[+neg]} [\text{PredP}[\varphi] [\text{down}]_j [\text{Pred}^0 [\text{VP requested that we } [\text{PolP } \%Pol^0_{[\%neg]} \text{ turn } t_j t_i]]]]]]]$

Crucially,  $W^0$  will be unable to attract VP out of  $\text{PredP}[\varphi]$ , which can be taken to result in a strong unchecked feature  $[+w]$ , and thus ungrammaticality. The same applies *mutatis mutandis* to (7a).<sup>22</sup> Likewise, the analysis of our German examples is straightforward, providing no further insights into the system developed in this section.

However, it remains to be seen to what extent this encoding of linearity/continuity effects in terms of the “freezing feature”  $\varphi$  is a satisfactory addition to this hybrid minimalist grammar. We think that its merits and defects can best be understood against the backdrop of a system that incorporates linear

<sup>22</sup> Obviously, the freezing effect induced by  $\varphi$  could be made more selective if that is required by empirical evidence.

precedence and interface conditions quite directly—that is, categorial grammar.

### 3.2 Forward Composition

It has already partly been shown and partly sketched by Steedman (1996, 2000b, 2003) how to provide a systematic and insightful account for the asymmetries we take to underlie CEST within his version of Combinatory Categorial Grammar (CCG). This crucially involves the Principle of Adjacency (Steedman 1996:5) according to which “[c]ombinatory rules may only apply to finitely many phonologically realized and string adjacent entities.” More specifically, continuous  $\sigma$ -regions correspond to CCG-constituents composing matrix and embedded predicate such that the string-final quantifier can scope over them. In cases like (7) the required composition is incomplete at the point where quantifier scope is determined. This results in a narrow-scope option only. Let us first, however, have a look at a wide-scope derivation of (1) as shown in (47).<sup>23</sup>

- (47)
- |     |                                                               |                       |
|-----|---------------------------------------------------------------|-----------------------|
| 1.  | she: S/(S\NP)                                                 |                       |
| 2.  | has_requested: (S\NP)/S                                       |                       |
| 3.  | that: S/S                                                     |                       |
| 4.  | they: S/(S\NP)                                                |                       |
| 5.  | read: (S\NP)/NP                                               |                       |
| 6.  | not_a_single_lb: S\((S\NP)                                    |                       |
| 7.  | she + has_requested: S/S                                      | C <sub>&gt;</sub> 1,2 |
| 8.  | she + has_requested + that: S/S                               | C <sub>&gt;</sub> 7,3 |
| 9.  | she + has_requested + that + they: S/(S\NP)                   | C <sub>&gt;</sub> 8,4 |
| 10. | she + has_requested + that + they + read: S/NP                | C <sub>&gt;</sub> 9,5 |
| 11. | she + has_requested + that + they + read + not_a_single_lb: S | E\ 10,6               |

(48) shows the semantic counterpart of the crucial function application step (47.11).

- (48)
- |    |                                                                                            |
|----|--------------------------------------------------------------------------------------------|
| a. | $\lambda P. \neg \exists y [LB(y) \wedge P(y)] (\lambda x. REQUESTED(she, READ(they, x)))$ |
| b. | $\neg \exists y [LB(y) \wedge REQUESTED(she, READ(they, y))]$                              |

For (7a) things are different. The subject quantifier has basically two options for combining with its syntactic neighbors. Rightward function application, which fixes narrow scope right away, is shown in (49).<sup>24</sup>

<sup>23</sup> Our notation follows Morrill (1994). C<sub>></sub>, denoting forward composition, is discussed in detail below. E/ is defined as follows. (i) *Slash Elimination* (E/):  $X/Y \ Y \Rightarrow_E X$ . Its leftward counterpart, E\, is defined the same way *mutatis mutandis*. Note the directionality asymmetry between subjects and objects, which, as it is reasonable to assume, is linked to (abstract) Case (see Steedman 2000b:45).

<sup>24</sup> Forward composition of Q<sub>-</sub> with the local verb equally results in narrow scope. Importantly, due to the inherent directionality difference between subjects and objects there is no leftward function application option for Q<sub>-</sub> taking the matrix predicate as argument.

- (49) a.  $\lambda P. \neg \exists y [\text{STUDENT}(y) \wedge P(y)] (\lambda x. \text{READ}(x, \text{our\_book}))$   
 b.  $\neg \exists y [\text{STUDENT}(y) \wedge \text{READ}(y, \text{our\_book})]$

The second option consists in forward composition affecting  $Q_{-}$  from the left. The syntactic side of this operation is provided in (50).

- (50) 1. she + has\_requested + that: S/S  
 2. not\_a\_single\_student: S/(S\NP)  
 3. she + has\_requested + that + not\_a\_single\_student: S/(S\NP) |  $C_{>}$  1,2

It might actually be less obvious why (50) does not give  $Q_{-}$  scope over the matrix predicate and, by transitivity, over the embedded one at a later stage as well. This, however, is prevented by the order inherent in function composition. (51) provides the full definition of Forward Composition, and (52) represents the semantic outcome of step (50.3).

- (51) *Forward Composition* ( $C_{>}$ ) (see Steedman 2000b:40)  
 $X/Y \quad Y/Z \Rightarrow_C X/Z$   
 $C \, fg \equiv \lambda x. f(gx)$

- (52) a.  $\lambda p. \text{REQUESTED}(\text{she}, p) \circ \lambda P. \neg \exists y [\text{STUDENT}(y) \wedge P(y)]$   
 b.  $\lambda Q [\lambda p. \text{REQUESTED}(\text{she}, p) (\lambda P. \neg \exists y [\text{STUDENT}(y) \wedge P(y)](Q))]$   
 c.  $\lambda Q [\lambda p. \text{REQUESTED}(\text{she}, p) (\neg \exists y [\text{STUDENT}(y) \wedge Q(y)])]$   
 d.  $\lambda Q. \text{REQUESTED}(\text{she}, \neg \exists y [\text{STUDENT}(y) \wedge Q(y)])$

To account for the intonational effects in (24) and (27)/(28), we have to enrich Steedman's system and employ the technique of "structural inhibition" discussed in Morrill 1994. Quite in the spirit of the approach there, we postulate that phonological boundaries, symbolized by %, are of type  $[\ ]^{\varphi}X/X$ —that is, trivial functions into "prosodically inhibited" categories. Additionally, we assume the following rule of  $[\ ]^{\varphi}$ -elimination.

- (53)  $[\ ]^{\varphi}\text{-Elimination}$  ( $E[\ ]^{\varphi}$ )  
 $[\ ]^{\varphi}X \Rightarrow_E X$

Assuming furthermore that there are no categories taking prosodically inhibited categories as arguments, we are left with (53) as the only way of getting rid of  $[\ ]^{\varphi}$ . Since this only works for a syntactically "complete" category  $X$ , narrow scope of a quantifier is enforced where it has to be built into  $X$  in order to make  $X$  "complete." The derivations in (54) and (55) show how the phonological boundaries scopally freeze  $Q_{-}$  inside of its local clause for (24) and (27), respectively.



- (54)
- |    |                                                                |                    |
|----|----------------------------------------------------------------|--------------------|
| 1. | has_requested: (S\NP)/S                                        |                    |
| 2. | %: [] <sup>o</sup> S/S                                         |                    |
| 3. | that + they + read: S/NP                                       |                    |
| 4. | not_a_single_lb: S\((S/NP)                                     |                    |
| 5. | that + they + read + not_a_single_lb: S                        | E\ 3,4             |
| 6. | % + that + they + read + not_a_single_lb: [] <sup>o</sup> S    | E/ 2,5             |
| 7. | % + that + they + read + not_a_single_lb: S                    | E[] <sup>o</sup> 6 |
| 8. | has_requested + % + that + they + read + not_a_single_lb: S\NP | E/ 1,7             |
- (55)
- |     |                                                                         |                    |
|-----|-------------------------------------------------------------------------|--------------------|
| 1.  | has_requested + that: (S\NP)/S                                          |                    |
| 2.  | the_students_who_finish_first: S/(S\NP)                                 |                    |
| 3.  | %: [] <sup>o</sup> (S\NP)/(S\NP)                                        |                    |
| 4.  | read: (S\NP)/NP                                                         |                    |
| 5.  | not_a_single_lb: T\((T/NP)                                              |                    |
| 6.  | read + not_a_single_lb: S\NP                                            | E\ 4,5             |
| 7.  | % + read + not_a_single_lb: [] <sup>o</sup> (S\NP)                      | E/ 3,6             |
| 8.  | % + read + not_a_single_lb: S\NP                                        | E[] <sup>o</sup> 7 |
| 9.  | the_students_..._first + % + read + not_a_single_lb: S                  | E/ 2,8             |
| 10. | has_requested + that + the_..._first + % + read + not_a_single_lb: S\NP | E/ 1,9             |

Crucially, given (53), []<sup>o</sup> can only be eliminated on a completed S in (54). There is no composition with the matrix predicate across % (\*C<sub>></sub>(54.1,54.2)). Thus, Q<sub>-</sub> has to combine with its local clause to yield a complete S and, consequently, Q<sub>-</sub> cannot take wide scope.

The same applies *mutatis mutandis* to (55) where it is a completed verb phrase (S\NP) that must be constructed.<sup>25</sup> Note that in (55.5) we have to use Steedman's "generalized" generalized quantifier type to allow backward function application (E\). Here the variable *T* ranges over "result types of functions over" NP (Steedman 2000b:44). Thus, since (S\NP)/NP is a function over NP, (S\NP) is in *T*'s range. For reasons of notational convenience, we use this generalization only where necessary.

For the cases involving phrasal verbs, (6) and (7b), it suffices to assume that, whereas the continuous *turn\_down* is of type (S\NP)/NP and therefore amenable to the treatment in (47), its discontinuous counterpart is analyzed as in (56).

- (56)
- |    |                                              |        |
|----|----------------------------------------------|--------|
| 1. | turn: V <sup>Prt</sup>                       |        |
| 2. | no_one: S\((S/NP)                            |        |
| 3. | down: ((S\NP)\V <sup>Prt</sup> )\((S\NP)/NP) |        |
| 4. | no_one + down: (S\NP)\V <sup>Prt</sup>       | E\ 2,3 |
| 5. | turn + no_one + down: S\NP                   | E\ 1,4 |

<sup>25</sup> Boundaries (%) must be taken to be "phrasal" in the appropriate sense, which seems to be a standard assumption in intonational phonology (see Ladd 1996). This prevents categories like the one in (i) from deriving the unwelcome wide-scope option for (27).

(i) %: []<sup>o</sup>((S\NP)/NP)/((S\NP)/NP)

Given this choice of categories,  $Q_{-}$  is linearly and derivationally “trapped” between verb and particle, which inevitably leads to narrow scope for  $Q_{-}$ .<sup>26</sup>

The contrast of (34) versus (31) has already been discussed by Steedman (2000b, 2003). Due to the OV nature of German, quantifiers in “extraposed” infinitivals like (31) are scopally restricted.

What we would like to add to this is our treatment of the intonational effect in (35b). The relevant derivation is given in (57). (*T* in (57.1) must be resolved to VP.)

- (57)
- |    |                                                          |            |     |
|----|----------------------------------------------------------|------------|-----|
| 1. | niemanden: $T/(T \backslash NP)$                         |            |     |
| 2. | zu_grüßen: $VP \backslash NP$                            |            |     |
| 3. | %: $[\ ]^0 VP \backslash VP$                             |            |     |
| 4. | versprach: $(S \backslash NP) \backslash VP$             |            |     |
| 5. | niemanden + zu_grüßen: VP                                | E/         | 1,2 |
| 6. | niemanden + zu_grüßen + %: $[\ ]^0 VP$                   | E \        | 5,3 |
| 7. | niemanden + zu_grüßen + %: VP                            | E $[\ ]^0$ | 6   |
| 8. | niemanden + zu_grüßen + % + versprach: $S \backslash NP$ | E \        | 7,4 |

Assuming in addition that *regelmäßig* (a) can be given type  $((S \backslash NP) \backslash VP) / ((S \backslash NP) \backslash VP)$  and (b) induces an intonational break on its left obligatorily, the narrow-scope effect in (40) is equally captured.<sup>27</sup>

<sup>26</sup> A combination of our technique with the wrapping account of Morrill (1994:113) might ultimately be more satisfactory. As it stands though, it is unclear how wrapping as such could prevent “late” combination with  $Q_{-}$ ; that is, after (i) has been formed via composition.

(i) has\_requested\_that\_we\_turn,down:  $(S \backslash NP) \uparrow NP$

In fact, this kind of technique has been used to model “quantifying-in” by Moortgat (1988) and Morrill (1994).

<sup>27</sup> A CCG-based approach to example (i) (repeated from fn. 18), which escapes the explanatory scope of CEST, requires decisions about an analysis of verb second (V2) that we are unwilling and unable to take here. See Steedman 2000b (sect. 6.7) for one approach.

(i) Sie meinte er liebe niemanden.  
 she thought he like.SUBJ no one  
 ‘She thought he loved no one.’

What is clear, though, is that, for the purpose at hand, V2 in German has to be taken as a rather superficial property that does not interfere with an essentially  $XP+V$  structure. Evidence for this is provided by (ii)—that is, the V2 counterpart of (34)—which seems to preserve a wide scope reading for  $Q_{-}$ .

(ii) Sie versprach niemanden zu grüßen.

We therefore notate V2-clauses as in (iii).

(iii) V2(sie+niemanden+zu\_grüßen+versprach)

V2 is an operator bringing about the necessary adjustments. Scope of  $Q_{-}$ , however, will be determined as before on the basis of the V-final structure; that is, the argument of V2. Second, it is well known that V2-complements are obligatorily extraposed (see Stechow & Sternefeld 1988). This will allow us to assimilate the account for the scope freezing effect in (i) to the account of (31) on the basis of representation (iv).

(iv) V2(sie+meinte+V2(er+niemanden+liebe))

Crucially,  $Q_{-}$  will be trapped inside its local clause, given that *meinte* and *liebe* cannot compose before  $Q_{-}$  is built in.

So far, we have seen the essentials of how to implement CEST within CCG.<sup>28</sup> However, the work by Steedman (2000a,b) would lead one to expect that the intonational boundaries we have appealed to induce information structural effects. The question thus arises whether the subcondition of CEST requiring prosodic continuity is reducible to information structural categories. That information structure plays an important role is in fact confirmed to the extent that the cases of wide scope taking in (1), (6), and (13a) are easiest to get if some kind of rise-fall pattern is imposed. (58) illustrates this for (13a).

(58) They have (/)FORced us to turn down NO(\) one.

This effect is familiar from the work on scope inversion by Büring (1997) and Krifka (1998). According to Büring's theory (58) carries the implicature that pairs of persons *x* and relations *R* comparable to forcing (e.g., suggesting or advising) are still "disputable" in terms of whether the group referred to by *they* *R*'s the group referred to by *us* to turn down *x*. Clearly, the material enclosed by the two accented categories in (58) must be interpreted as background. However, it is unclear which information structural dimension distinguishes (1) from (27). Both *they* and *the students who finish first* can be interpreted as background. Nevertheless the boundary induced by intonational phrasing in the case of (27) interferes with wide-scope construal of *Q*<sub>1</sub>.

At this stage we can only add some speculative remarks on where to look for the missing extra dimension in developing a deeper account for the prosodic part of CEST. It seems to us that our "σ-region" has to undergo some kind of (pre-)restructuring, the latter an incomplete form of complex predicate formation. Extended scope of negation is one of the well-known hallmarks of restructuring (see Haegeman & van Riemsdijk 1986, among many others). Further factors known to be involved are canonical directionality, adjacency, and the lexical content of the matrix predicate. Examples (1), (6), and (13a) are surprising from this perspective to the extent that adjacency between matrix and embedded predicate is not strict. However, it is significant, we believe, that the examples typically discussed in the literature involve intervening personal pronouns—that is, weak function words that prosodically integrate into one of their neighbors.<sup>29</sup> They therefore do not appear to block (pre-)restructuring—in other words, they are compatible with "weak" adjacency.

The upshot of this is that the CCG-based scoping account may have to be supplemented with a mechanism of *R*-feature [*R* = restructuring] spreading

<sup>28</sup> For the "raising-to-object" analysis of ECM cases like (30), types could be assigned as in (i).

(i) a. expected: ((S\NP)/VP)/NP  
b. no\_one: *T*\(*T*/NP)

<sup>29</sup> Functional items like *that* and *to*, which are prosodically weak as well, can be treated as formal predicate parts.

that constrains function composition. Reluctantly, we leave the exact detail of this mechanism for further research.<sup>30</sup>

#### 4. Conclusion

In this paper we have shown that a number of asymmetries concerning the extended scope taking of negative quantifiers in English and German are subject to a condition on the linear and prosodic shape of the domain the quantifier scopes over, the CEST:

##### (25) *Condition on Extended Scope Taking (CEST)*

Extending the scope of a negative quantifier  $Q \neg$  over a region  $\sigma$  requires  $\sigma$  to be linearly and prosodically continuous.

We then demonstrated how to implement CEST in a Y-model preserving minimalist extension of Kayne (1998) and an extension of CCG (Steedman 1996, 2000a,b) the latter an explicitly multidimensional alternative to the Y-model. For our concerns, the most important difference between these systems is the status of linear notions like adjacency and directionality. These play an immediate role in CCG, allowing an alternative to scope taking by movement in terms of function composition. They are equally instrumental in implementing the homogeneity conditions underlying CEST.

For the syntactic part of the minimalist alternative, these linear notions are epiphenomenal. Instead, one has to employ and constrain a number of additional movement triggers and operations.

Both systems are crucially supplemented by an “inhibitory” or “freezing” operator  $\phi$ . From a very abstract perspective it can be said that where the global minimalist freezing operator  $\phi$  does not allow constituents to be taken apart any further, directionality requirements together with  $\phi$ ’s categorial counterpart do not allow certain things to be put together. It remains to be seen how natural these additions fit into the systems discussed. Our investigation here is only a first step in that direction.<sup>31</sup>

<sup>30</sup> An additional—more controversial—application of CEST could be providing the conditions on Q-shift in Icelandic. Svenonius (2000:267, 270) reports the following subject-object asymmetry.

- (i) a. %Hún hafði margt<sub>i</sub> viljað að hann gætur keypt  $t_i$ .  
           she had many wanted that he could.SUBJ bought  
           ‘She had wanted him to be able to buy many.’
- b. \*Hún hafði marga stelpur<sub>i</sub> viljað að  $t_i$  kæmu í veisluna.  
           she had many girls wanted that come.SUBJ to the.party  
           ‘She had wanted that many girls come to the party.’

The only assumption we have to make is that (certain) traces function like prosodic boundaries in blocking composition of the  $\sigma$ -region (or R-feature spreading). This would involve a weakening of Steedman’s Adjacency Principle, quoted earlier with respect to the condition on “phonological realization.”

<sup>31</sup> For a thorough discussion of such techniques in categorial Type Logical Grammar, see Morrill 1994.

We hope, however, that the question as to whether such different formalizations will ultimately be a matter of taste can be better understood on the basis of our study that shows both rivaling systems at work in a nicely confined but nontrivial domain.

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